CO - 1 AT - 1
DSA0311 - NLP

1. Resume Information Extraction

Code :

```
 import re

def extract_resume(text):
    name = re.search(r"Name:\s*([A-Za-z ]+)", text)
    email = re.findall(r'[\w.-]+@[\w.-]+\.\w+', text)
    mobile = re.findall(r'\b\d{10}\b', text)
    skills = re.findall(
        r'Python|Java|SQL|Machine Learning|NLP',
        text, re.I
    )
    exp = re.search(r'(\d+)\s*years?', text, re.I)

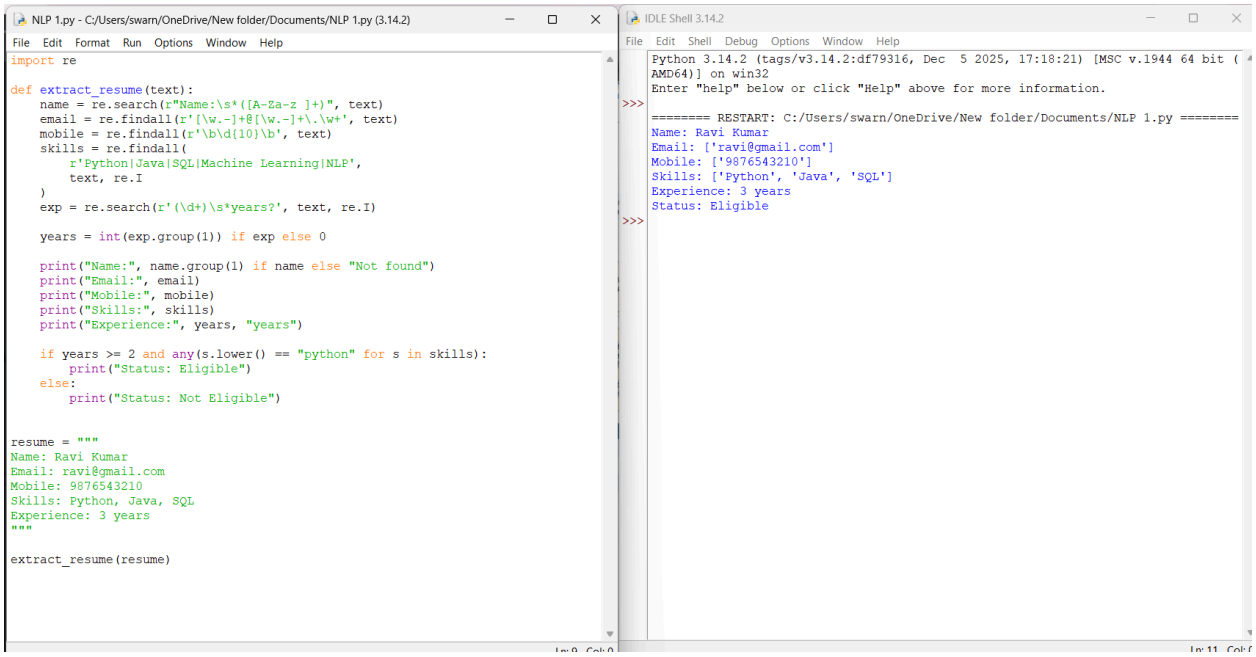
    years = int(exp.group(1)) if exp else 0

    print("Name:", name.group(1) if name else "Not found")
    print("Email:", email)
    print("Mobile:", mobile)
    print("Skills:", skills)
    print("Experience:", years, "years")

    if years >= 2 and any(s.lower() == "python" for s in skills):
        print("Status: Eligible")
    else:
        print("Status: Not Eligible")


resume = """
Name: Ravi Kumar
Email: ravi@gmail.com
Mobile: 9876543210
Skills: Python, Java, SQL
Experience: 3 years
"""

extract_resume(resume)
```



2. Product Search System

Code:

```
import re

products = [
    "Python Book",
    "Java Book",
    "Python Course",
    "SQL Guide",
    "Machine Learning Course"
]

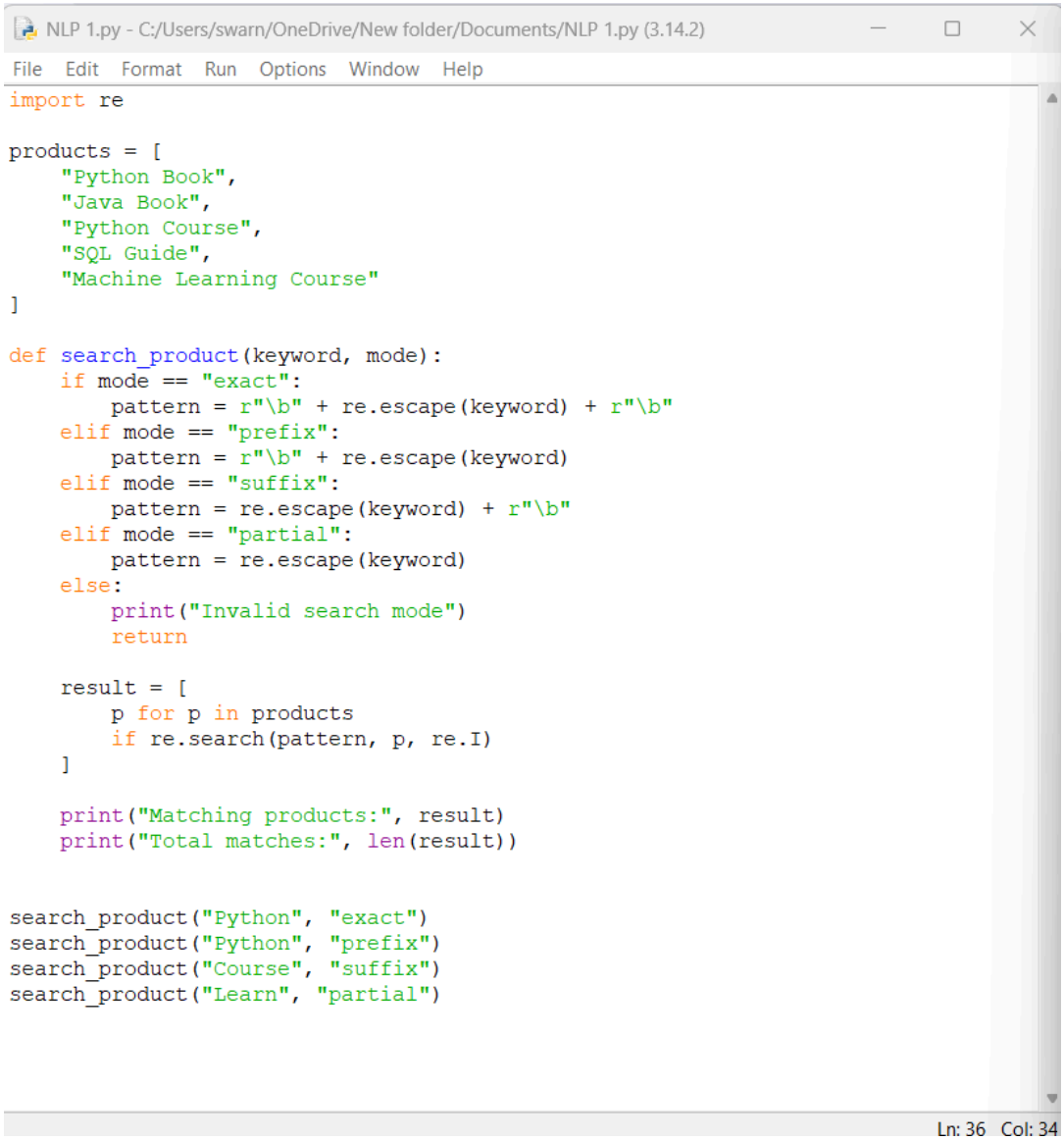
def search_product(keyword, mode):
    if mode == "exact":
        pattern = r"\b" + re.escape(keyword) + r"\b"
    elif mode == "prefix":
        pattern = r"\b" + re.escape(keyword)
    elif mode == "suffix":
        pattern = re.escape(keyword) + r"\b"
    elif mode == "partial":
        pattern = re.escape(keyword)
    else:
        print("Invalid search mode")
        return

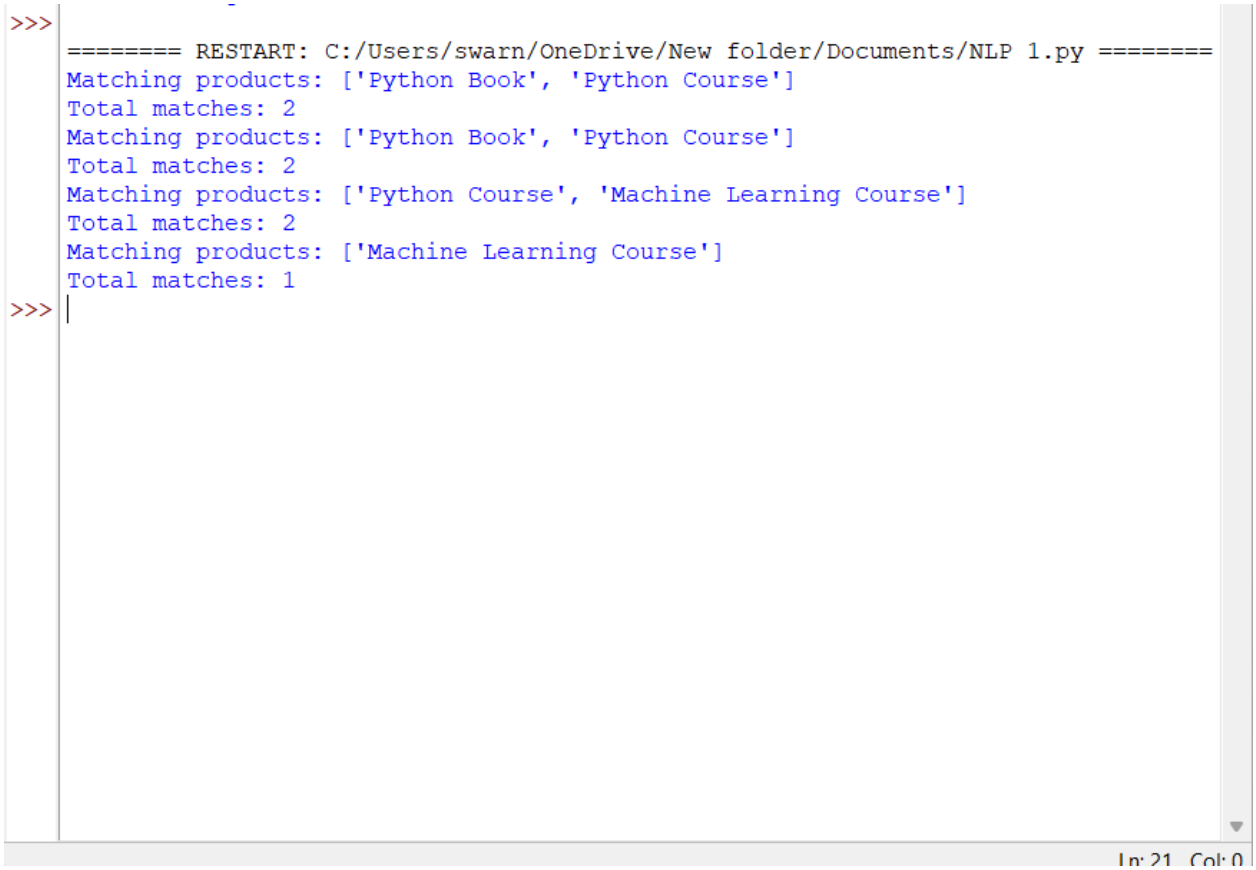
    result = [
        p for p in products
```

```
        if re.search(pattern, p, re.I)
    ]

    print("Matching products:", result)
    print("Total matches:", len(result))


search_product("Python", "exact")
search_product("Python", "prefix")
search_product("Course", "suffix")
search_product("Learn", "partial")
```





3. University Registration System

Code:

```
import re

def validate(pattern, value):
    return bool(re.fullmatch(pattern, value))

reg_no = input("Register number: ")
email = input("Institutional email: ")
course = input("Course code: ")
semester = input("Semester (1-8): ")
mobile = input("Mobile number: ")

checks = {
    "Register Number": validate(r"\d{12}", reg_no),
    "Email": validate(r"\w+@college\.edu", email),
    "Course Code": validate(r"[A-Z]{2,4}\d{3}", course),
    "Semester": validate(r"[1-8]", semester),
    "Mobile Number": validate(r"[6-9]\d{9}", mobile)
}

for field, valid in checks.items():
```

```
    print(field + ":", "Valid" if valid else "Invalid")

if all(checks.values()):
    print("Registration Successful")
else:
    print("Registration Failed")
```

